

LECTURE 8

Reinforcement Learning Part 3: Multi-Agent Systems & Autonomous Combat Platforms

MARL • CTDE • QMIX • MAPPO • MADDPG • credit assignment • emergent tactics

Case study: 6-unit combined arms (3 ground + 2 drones + 1 SEAD) — emergent flanking and decoy tactics

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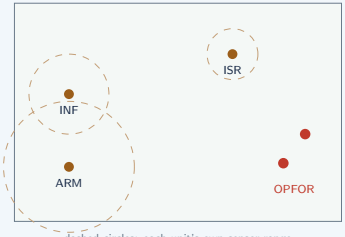
AI & Machine Learning

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Why Multi-Agent RL? — The Limits of Single-Agent Approaches

What a single agent cannot model

① **Simultaneous actors:** 6 units act concurrently; each action affects everyone's reward; one agent controls one entity.
② **Partial observability per unit:** a drone sees 200 m, a tank 500 m; nobody has the full picture.
③ **Cooperative reward:** mission success is a JOINT outcome; individual credit is ambiguous.
④ **Emergent tactics:** flanking needs unit A to distract while B moves.
⑤ **Adversarial dynamics:** OPFOR is also multi-agent and adapts — the environment is non-stationary.



dashed circles: each unit's own sensor range

No unit sees the whole battlefield, and none can win alone — the defining features of a multi-agent problem.

The MARL Paradigm — N Learning Agents in One Shared World

DEC-POMDP

$$\big(S, \{A_i\}, P, \{r_i\}, \{\Omega_i\}, O, \gamma\big)$$

$$\pi_{\theta_i}(a_i|o_i), \quad \mathbf{a} = (a_1, \ldots, a_N), \quad r(s, \mathbf{a}) \text{ or } r_i(s, \mathbf{a})$$

Fundamental challenge — non-stationarity: from agent i's view, the other $N-1$ agents are part of the environment AND they are learning. The "environment" keeps changing even with fixed physics, so single-agent convergence theorems no longer apply.

Fully cooperative
shared reward
$r_i = r$
combined-arms attack

Fully competitive
zero-sum
Blue vs Red

Mixed
cooperative Blue team vs Red team

most real military scenarios

The mark of modern warfare: unmanned systems fighting in coordinated swarms, autonomous wingmen, manned–unmanned teams — all multi-agent problems.

CTDE — Centralised Training, Decentralised Execution



- **Training:** use GLOBAL information (all observations, actions, full state) to train a centralised critic/mixer $\Rightarrow$ better credit assignment; coordination emerges
- **Execution:** each agent acts on its OWN local observation o_i $\Rightarrow$ no bandwidth needed, works even when comms are jammed

DECOMPOSITION

actor: $\pi_i(a_i|o_i;\theta_i)$ critic: $V(s, o_{1..N}, a_{1..N};\phi)$

Military analogy: the commander briefs each unit on the plan (centralised), then radio silence — units execute independently on what they see (decentralised). This is mission command: shared intent, local initiative.

Three CTDE Algorithms — QMIX, MAPPO and MADDPG

QMIX (Rashid 2018) — value-based

Each agent has a local $Q_i(o_i, a_i)$. A mixer combines them: $Q_{\text{tot}} = f(Q_1, \ldots, Q_N; s)$.

Key constraint — monotonicity: $\partial Q_{\text{tot}}/\partial Q_i > 0$ (positive weights) $\Rightarrow$ the joint argmax equals the per-agent argmaxes $\Rightarrow$ decentralised execution guaranteed.

Training: TD loss on Q_{tot} against $r + \gamma Q_{\text{tot}}^{\text{target}}$.

MAPPO (Yu 2022) — policy gradient

Extends PPO to N agents with **parameter sharing** across agents.

Centralised critic $V(s)$ on the full global state; each agent's PPO update uses $V(s)$ as baseline.

Works well when agents are homogeneous; sharing dramatically reduces parameter count.

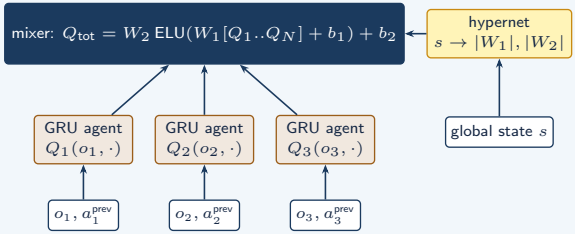
MADDPG (Lowe 2017) — actor–critic

Deterministic actors $a_i = \mu_i(o_i; \theta_i)$.

Each agent has its OWN centralised critic $Q_i(s, a_1, \ldots, a_N; \phi_i)$ that sees every agent's action.

Suits continuous actions and mixed cooperative–competitive settings (each agent can have its own reward).

QMIX — Global Q as a Monotonic Mix of Local Q-Values



- Agent networks: one per agent, shared weights; a GRU handles partial observability
- The hypernetwork maps s to mixing weights, made positive with absolute values; biases may take any sign

MONOTONICITY ⇒ DECENTRALISED GREEDY EXECUTION

$$\frac{\partial Q_{\text{tot}}}{\partial Q_i} > 0 \;\Rightarrow$$

$$\arg\max_{\mathbf{a}} Q_{\text{tot}} = \left(\arg\max_{a_1} Q_1, \ldots, \arg\max_{a_N} Q_N\right)$$

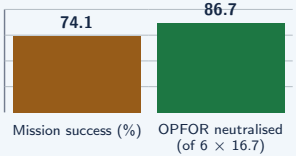
Each agent independently picks ITS best action — and the team's joint choice is still optimal for Q_{tot}. No negotiation needed at execution time.

Limitation: monotonic mixing cannot represent situations where one agent's best action depends on another's (e.g. "only advance if the other unit holds"). Richer coordination needs other methods.

QMIX Case Study — Combined Arms with 6 Agents

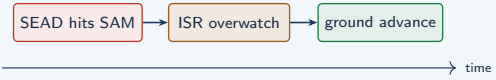
Environment

3 ground units (infantry, armour, artillery) + 2 ISR drones + 1 SEAD aircraft. Objective: destroy a defended position held by 2 SAM sites and 4 OPFOR infantry. Observation per agent (42-D): own position, health, ammo, role; visible units in sensor range; friendly positions if comms are not jammed. QMIX state = $6 \times 42 = 252$-D.



QMIX after 200K episodes: success 74.1%; 38.2 turns per episode; 1.8 friendly casualties; 5.2/6 targets neutralised; ∼18 h on 8 CPU workers.

Emergent behaviour: the SEAD aircraft ALWAYS neutralises the SAM sites first, and ground units converge only after the SAMs are down. The agents discovered suppression-then-advance — standard SEAD doctrine — from reward alone.



learned sequence: suppress, observe, then advance

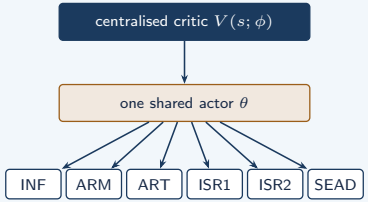
MAPPO — Multi-Agent PPO with a Shared Centralised Critic

Four design choices

① **Shared actor parameters:** all agents use one network $\pi(a_i|o_i;\theta)$ with agent ID one-hot in the input: 6 agents $\rightarrow$ 1 actor ($N\times$ efficiency).
② **Centralised critic:** $V(s;\phi)$ on the global state, one critic for all; $\hat{A}_i = r + \gamma V(s') - V(s)$.
③ **PPO clipping per agent:** L_i^{CLIP} with $r_i(\theta) = \pi_i^{\text{new}}/\pi_i^{\text{old}}$.
④ **Advantage normalisation:** $\hat{A}_i \leftarrow (\hat{A}_i - \text{mean})/(\text{std} + \epsilon)$ across agents and steps — critical for stability.

MAPPO LOSS

$$L = \frac{1}{N}\sum_i L_i^{\text{CLIP}}(\theta) + c_1 L_{\text{critic}}(\phi) - c_2 H(\pi_i)$$

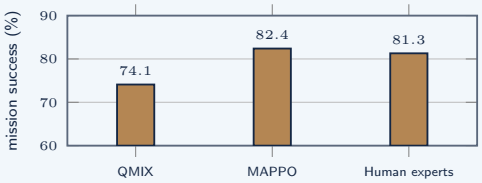


Parameter sharing means every unit's experience trains the same network: the infantry agent benefits from what the armour agent learned, as units share lessons in an after-action review.

MAPPO vs QMIX — and the Human-Expert Benchmark

QMIX	MAPPO
monotonic coordination guarantee	PPO clipping: robust
off-policy (replay, sample-efficient)	on-policy (fresh data)
purely cooperative (shared reward)	mixed coop.–competitive
homogeneous obs. preferred	heterogeneous agents OK

Combined-arms results



MAPPO: mission success **82.4%** (best no-comms MARL result); 33.7 turns per episode (faster than QMIX); 1.2 casualties per mission; 5.7/6 targets neutralised.

Human-expert team benchmark

3 experienced military planners, same scenario, manual decisions each turn: **81.3%** success, 41.8 turns, 1.6 casualties. **MAPPO slightly exceeds the human team** — in this simulation.

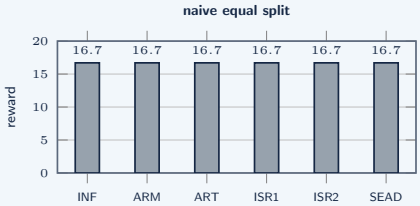
82.4% vs 81.3% is a 1.1-point gap: without the number of evaluation episodes and a confidence interval we cannot call it a real difference (Section 0 of Lecture 1). The honest claim is "comparable to experts".

Credit Assignment — Who Deserves Credit for a Joint Victory?

Mission succeeds (+100). Who earned it?

Agent 1 infantry: suppressive fire. Agent 2 armour: advanced under cover. Agent 3 artillery: fired too early — missed. Agent 4 drone 1: ISR (critical). Agent 5 drone 2: malfunctioned, did nothing. Agent 6 SEAD: neutralised the SAM (critical).

Naive split: every agent gets $100/6 = +16.7$ — the artillery that missed and the drone that did nothing are REWARDED. Agents learn "do nothing, still win": the **free-rider problem**.



QMIX's partial answer: the factorisation gives each agent its own Q_i; an idle agent's $Q_5 \approx 0$, and the mixer weights contributions by the global state.

COMA — Counterfactual Credit: "What If Only I Had Acted Differently?"

COUNTERFACTUAL BASELINE (FOERSTER 2018)

$$b_i(s, \mathbf{a}_{-i}) = \sum_{a_i'} \pi_i(a_i' | o_i)\, Q\big(s, (a_i', \mathbf{a}_{-i})\big)$$

$$A_i(s, \mathbf{a}) = Q(s, \mathbf{a}) - b_i(s, \mathbf{a}_{-i})$$

- Centralised critic $Q(s, a_1, \ldots, a_N)$
- "What would the team's value be if everyone else did the same but agent i acted on average?"
- A_i = how much a_i improved over agent i's own average

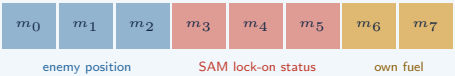


The missed artillery shot (-3.2) and the idle drone (-1.1) are correctly penalised; armour ($+12.1$), SEAD ($+11.4$) and ISR ($+9.8$) earn the most credit.

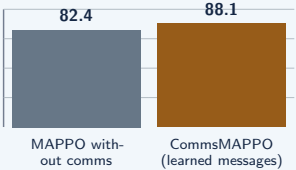
The PPTX states that the advantages sum to ≈ 0 (a baseline property); the six values shown sum to $+37.4$, so that claim does not hold for this table.

Set-up

Add a channel: each agent broadcasts a message $m_i \in \mathbb{R}^8$ (a learned "language"). Actor input: $[o_i, m_1, \ldots, m_{N-1}]$. Agents learn both WHAT to broadcast and how to USE others' messages — purely by gradient descent.



Linear-probe analysis of drone messages after 50K training episodes.

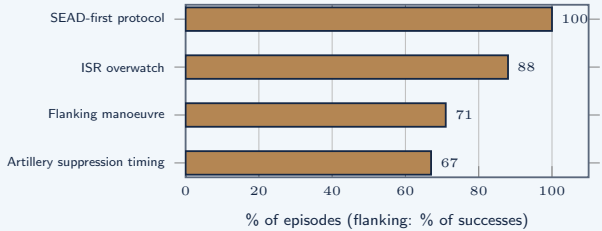


Mission success (%).

Agents spontaneously developed a code carrying tactically critical information — enemy positions seen by the drones, SAM lock-on threat, and fuel state — without anyone defining the vocabulary.

A learned code is opaque to humans and could be spoofed by an adversary who learns it. It must be decoded (probes) and secured before any use.

Top emergent tactics (identified by domain experts in replay logs)



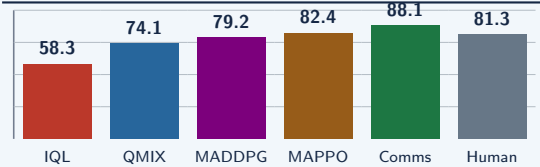
A fifth tactic, bounding advance, was also identified (frequency not reported).

- **SEAD-first (100%):** SAM is the existential threat; any episode with an active SAM fails often
- **ISR overwatch (88%):** drones take high-ground vantage points covering ground units' blind spots — learned geometric coverage
- **Flanking (71% of successes):** infantry distracts frontally while armour flanks — never explicitly rewarded; it emerged from the joint value gradient
- **Artillery timing (67%):** fires during the flanking move

Implication for doctrine: MARL rediscovers valid doctrine and may reveal NEW coordination patterns; expert review of replay logs is itself a source of tactical insight.

Algorithm Comparison — All MARL Methods vs the Human Expert Team

Method	Type	Success	Casualties	Turns	Train eps	Key strength
IQL (independent Q)	single-agent×6	58.3%	2.8	44.1	100K	baseline; ignores structure
QMIX	value CTDE	74.1%	1.8	38.2	200K	monotonic mix; greedy decentralised
MADDPG	actor–critic CTDE	79.2%	1.6	35.8	250K	per-agent critics; continuous
MAPPO	PG CTDE	82.4%	1.2	33.7	150K	shared params; best without comms
CommsMAPPO	PG + comms	**88.1%**	**0.9**	**29.4**	180K	emergent language
Human expert team	human	81.3%	1.6	41.8	—	benchmark (with comms)



Takeaways: ignoring multi-agent structure (IQL, 58.3%) is deeply costly; MAPPO matches or slightly exceeds the human team; learned communication adds a further ∼6 points.

When to Use Which — and a MARL Safety Warning

QMIX

Purely cooperative; discrete actions; communication-denied; interpretable per-agent Q-values; off-policy sample efficiency. (Note: MAPPO still scored higher without comms in this study.)

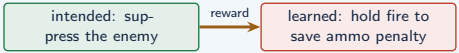
MAPPO

Large teams; heterogeneous agents; mixed tasks — the most robust choice overall.

MADDPG

Continuous actions; mixed competitive settings.

MARL safety warning — emergent behaviours are UNPREDICTABLE. In some runs agents learned to conserve ammunition by NOT FIRING: locally rational under the reward, tactically bad. Deployment needs extensive red-teaming.



the agent optimises what you reward, not what you meant

Autonomous Combat Platforms — Levels of Autonomy

Level 4 — fully autonomous
seeks; decides; engages for hours or days; prohibited in most militaries

Level 3 — human-out-of-the-loop (LAWS)
engages without per-action approval; contested under IHL

Level 2 — human-on-the-loop
AI nominates; human approves each action

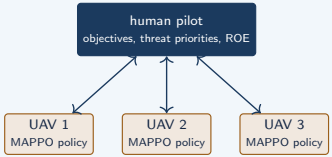
Level 1 — human-operated
full human control; AI assists targeting

Illustrative ladder after DoD / UK MoD style frameworks.

Examples

L1: drone with AI camera assist. L2: **MAPPO advisory mode** — 6 agents each recommend an action and a human approves (this course). L3: a loitering munition with target criteria; or QMIX in a comms-contested environment. L4: long-duration independent missions.

Level 2 is practical, legally reviewable and deployable now. Every step up the ladder transfers judgement from accountable humans to a policy that was optimised in simulation.



datalink $\leq$2 s latency; policy pre-trained on the ground, updated overnight

ROE as reward shaping

The human provides waypoints, threat priorities and ROE; MAPPO encodes them as penalties: r_{ROE} for engaging non-combatants, r_{prop} for disproportionate engagement. Agents optimise the mission inside these constraints.

Open challenges

① trust calibration: when to override the agent; ② adversarial robustness: OPFOR exploiting learned patterns; ③ accountability: who is responsible for an AI action?; ④ sim-to-real transfer.

Ethical bright line: MARL for ADVISORY and PLANNING purposes is current best practice. Autonomous lethal engagement (Level 3+) needs legal clarity under IHL on meaningful human control. This course treats MARL as decision support — **the human remains the decision-maker for lethal force.**

A penalty term is not a legal safeguard: reward shaping makes violations *less likely*, not impossible. Human approval provides the guarantee.

Coordination without a central controller is everywhere in nature, engineering and war

BIOLOGY
Ants coordinate through pheromone trails left in the environment (stigmergy, Grassé 1959); ant colony optimisation (Dorigo, 1992) borrows the idea.
Defence: decentralised route planning for swarms that cannot talk directly.

STATISTICS / GAME THEORY
Nash equilibrium (1950); Axelrod's iterated prisoner's-dilemma tournaments (1980s) showed tit-for-tat cooperation emerging between self-interested players.
Defence: deterrence, escalation and coalition behaviour in strategic wargames.

PSYCHOLOGY
Social loafing: individuals pull less hard on a rope when pulling in a group (Ringelmann; Latáne et al., 1979) — the human free-rider problem.
MARL twin: equal reward splits teach agents to coast; COMA-style credit fixes it.

AERONAUTICS
TCAS coordinates two aircraft (one climbs, one descends). At Überlingen (2002) one crew followed a controller's contrary instruction and the aircraft collided.
Defence: in MUM-T, authority between human and machine must be unambiguous.

MILITARY DOCTRINE
Mission command (*Auftragstaktik*): commanders share intent centrally; subordinates act on local judgement when communications fail.
MARL twin: this is CTDE — centralised training, decentralised execution.

GAMES & WARGAMING
AlphaStar reached Grandmaster level in StarCraft II by training a *league* of agents against each other (Vinyals et al., *Nature*, 2019) to avoid exploitable strategies.
Use: league self-play for robust OPFOR agents.

Common thread: shared intent plus local action scales; unclear credit or authority breaks teams of people and agents alike.

Lecture 8 Summary — From CTDE to Emergent Combined-Arms Tactics

DEC-POMDP & MARL
N agents, partial observations, joint reward; non-stationarity is the core challenge; mixed coop.–competitive is the norm.
Foundation

CTDE
Train centralised, execute decentralised; plan rehearsal → radio silence; coordination without field comms.
Dominant defence paradigm

QMIX
Monotonic Q_{tot}; decentralised greedy execution; 74.1%; off-policy, interpretable.
Use: cooperative, comms-denied

MAPPO
Shared actor + centralised critic; 82.4% vs human team 81.3%; PPO stability.
Go-to for cooperative MARL

EMERGENT TACTICS
SEAD-first, ISR overwatch, flanking, artillery timing — all from reward; plus an emergent message code.
Validates doctrine; reveals patterns

MUM-T & LAWS ETHICS
Level 2 advisory is best practice; ROE as reward shaping; lethal autonomy unresolved under IHL.
Human decides on lethal force

Lecture 9: putting the whole course together

- Full course concept map
- Cross-lecture case-study threads revisited
- Integrated AI/ML pipeline for defence scenarios
- Common exam misconceptions
- Rapid-fire MCQ revision
- Key formula sheet

Assignment 8 is released with this pack. **Lecture 10:** student project presentations and the final Q&A panel.

